

An Eye-tracked XR Visual Deficit Simulation Suite for Ocular Disease Education and Assisted Diagnosis

Jason Orlosky*
Augusta University

Tommy Bui
Augusta University

Nicole Winston
Augusta University

Wanda Jirau-Rosalay
Augusta University

Shilpa Brown
Augusta University

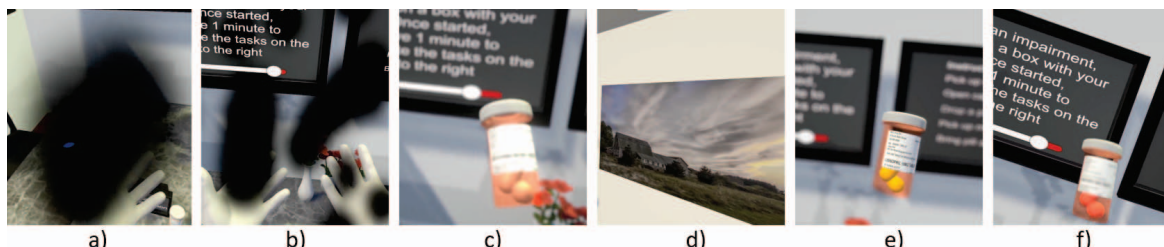


Figure 1: Images of the VR interface for the visual impairment suite showing the following simulated deficits, including a) macular degeneration, b) diabetic retinopathy, c) cataracts, d) colorblindness, e) myopia, and f) hyperopia.

ABSTRACT

In this demo, we present a suite of visual deficits implemented in both augmented and virtual reality to assist with education and diagnosis of ocular disorders, diseases, and deficits. More specifically, we use eye tracking to better simulate the difficulties in diseases such as macular degeneration, diabetic retinopathy, retinal tears, and other gaze-dependent deficits. We make use of both virtual reality (VR) and video see-through augmented reality (VST AR) to give users a unique perspective of life through the eyes of patients.

1 INTRODUCTION AND PRIOR WORK

In health care settings, caregivers and healthcare professionals often have difficulty understanding what a patient is going through. This is particularly true for patients with visual deficits since replicating those deficits for a caregivers and healthcare professionals in a way that lets them experience the difficulty of day-to-day tasks. To address these problems, we have designed a suite of simulated visual deficits that can be used to enhance student learning and provide training to physicians and caregivers. To achieve this, we built an eye-tracked Virtual Reality (VR) environment which we can simulate a multitude of different visual deficits and diseases.

Though empathy training and working with patient actors can help alleviate this problem, there continues to be multiple challenges in training physicians and medical students, and the literature tends to emphasize behaviors that enhance perceived empathy from the patient's perspective. In Patel et al.'s systematic review regarding empathy-related curricula, they mention five physician-oriented behaviors that improve empathy that patients perceive, only one of which mentions a simulated experience [4]. Other simulated visual deficit systems are available, but have not taken advantage of the XR features available today. For example, Lewis et al. simulated a 3D visual deficit environment, but this was not stereoscopic or world-tracked [3]. Additional deficits were simulated on a static screen later by Stewart et al. [5]. Video See-through (VST) simulations were developed as early as 2015, but these were not eye tracked [2]. Our work seeks to close these technology gaps and broaden the range of available deficits.

*e-mail: jorlosky@augusta.edu

2 SIMULATION SUITE ENVIRONMENT AND DEFICITS

Our hardware environment includes a PC with a core i7 processor and NVIDIA RTX 3080. The display for VR rendering we used was an HTC Vive Pro Eye, and we used a Meta Quest Pro for Augmented Reality (AR). The virtual environment was implemented using Unity 3D 2020.3.21.f1. The virtual environment itself is set in a room designed to resemble an office or kitchen with a table in the middle, as shown in Figure X. On this table are a series of buttons, as shown in Figure 1. Once depressed, each of these buttons starts a respective simulation of that visual deficit. In addition, this triggers the start of a timer that takes the patient through a self-medication task based on the scenario described in [1]. The AR environment makes use of the Meta Quest Pro's video see-through texture with the world tracking disabled (to prevent the virtual fence from appearing) and uses the integrated eye tracking API in the same way as the Vive Pro Eye.

2.1 Deficits in Virtual Reality: Medication Task

The deficits we implemented for VR are part of a self-medication task in which the user selects one of the six deficits in Figure 1.

2.1.1 Macular Degeneration

Macular degeneration is the most common cause of irreversible blindness in the elderly in developed countries characterized by a dark spot or loss of a spot of vision, typically in the middle of the field of view (FoV). Some individuals with this deficit describe the region as having surrounding colors fill the area. The deficit stems from degeneration of the macula, an area in the center of the retina that specializes in high acuity of vision. Age related macular degeneration is progressive, eye-dependent, and as the eye moves, the deficits move to the center of vision. We have this implemented for both AR and VR, shown in a) of Figure 1 and c) of Figure 2¹, respectively, which consists of a non-uniform texture that decreases in opacity as distance from the center of the field of view (FoV) increases. The texture is attached to the gaze ray of the user at a fixed distance so that it moves with their eye movements.

2.1.2 Diabetic Retinopathy

Diabetic Retinopathy is a common cause of blindness that usually presents after many years of poorly controlled diabetes. Diabetic retinopathy is progressive and similar to macular degeneration but may be asymptomatic or in some cases characterised by multiple, non-uniform spots in the field of vision. These can vary greatly

¹Also available in our video: <https://www.youtube.com/watch?v=9kRiVoQxgl>

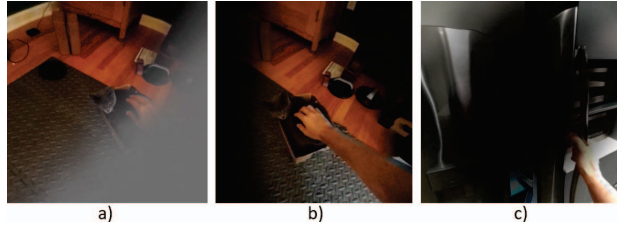


Figure 2: Images showing three of the eye-tracked deficits we implemented in video see-through AR, including a) gray curtain, b) side shadows, and c) macular degeneration.

in size and shape and are also generally unique to each eye. To implement this, we created a texture similar to macular degeneration, but that took up a larger portion of the FoV and contained multiple spots. This texture is similarly affixed to the user's gaze-ray, though it covers a majority of the visual field.

2.1.3 Color Blindness

Color blindness is when individuals have difficulty distinguishing between certain colors. It often results in a limited ability to perceive specific hues (red-green, blue-yellow, and complete color deficiency). While color blindness itself does not worsen over time, its impact on daily tasks and activities can be ongoing, emphasizing the need for adaptive strategies. To implement this, we use a Unity Post-process Volume, which allows for shading of the scene camera on a per-eye basis. We then enable the color grading feature, which allows for modification of the individual red, green, and blue components that control color components in the entire scene. These are reduced to produce the effect shown in c) of Figure 1.

2.1.4 Myopia and Hyperopia

Myopia, commonly known as nearsightedness, is a refractive error that causes distant objects to appear blurry while close objects remain clear. Individuals with myopia often describe difficulty seeing objects at a distance, depending on the severity of myopia. The deficit is primarily caused by the length of the eye being too long or the cornea having excessive curvature, preventing optimal convergence of light onto the retina. Myopia tends to develop during childhood and adolescence, and its severity can change over time, often stabilizing in adulthood.

Hyperopia, or farsightedness, is a refractive error where distant objects are seen more clearly than close ones. The deficit results from the eyeball being too short or the cornea having insufficient curvature. Due to the ability for the intraocular lens to accommodate, hyperopia can compensate depending on severity, however, aging (presbyopia) and high magnitude of hyperopia can prevent optimal convergence of light on the retina. Hyperopia can change over time, with some individuals experiencing an increase in symptoms as they age (presbyopia). Like myopia, hyperopia can be corrected with glasses, contact lenses, or refractive surgery.

Both myopia and hyperopia are implemented with the depth of field feature of the post-processing volume. This allows us to set focal distance, aperture, focal length and blur size, which are focus distance = 28cm, aperture = 6.1, and blur size = large for myopia and focus distance = 3.69m and aperture = 6.9 for hyperopia.

2.1.5 Cataracts

Cataracts are usually age related clouding or discoloration of the eye's intra-ocular lens, resulting in a slowly progressive decrease in visual clarity. Individuals with cataracts often report blurred or cloudy vision, increased glare, and difficulty seeing in the dark. The deficit arises from opacification and discoloration of the lens. Cataracts occur per-eye and progression depends on the underlying etiology of the lens opacification. Cataracts were also implemented

with the post-processing volume using the vignette (intensity = 0.65 and smoothness = 0.394), bloom (intensity = 8 and threshold = .72, and depth of field (focus distance = 0.75, aperture = 1.3, focal length = 1, and blur size = small) features.

2.2 Deficits in VST Augmented Reality: Retinal Tears

The deficits in this section are eye-tracked and implemented in the Meta Quest Pro. The interface is completely mobile, giving the user the deficit and letting them interact with the real world.

2.2.1 Side Shadows

Side shadows from retinal tears refer to the perception of a shadow or dark area in one's peripheral vision. This can occur when there is a disruption in the normal. Individuals with this deficit may describe a partial or complete shadowing effect, affecting their ability to see objects in the affected area that stay constant regardless of gaze position. We achieve this effect by placing a dark texture on either side of the user's periphery. Each texture increases in opacity as distance into the periphery increases, with the furthestmost regions being completely opaque.

2.2.2 Gray Curtain

Curtains resulting from retinal tears describe a visual disturbance where individuals perceive a gray or dark curtain descending over their visual field. This visual deficit is indicative of detached retina, and individuals may report a sudden onset of the gray curtain, accompanied by a rapid decline in vision. Our implementation is similar to the side shadows and macular degeneration mentioned previously, which includes a gray texture that spans the central field of view to the right side of the display, extending into the entire periphery. The center is semi-transparent, with a gradual gradient that increases in opacity farther into the periphery.

3 DISCUSSION AND CONCLUSION

In this work, we presented a simulation suite of visual deficits to help with the understanding the impact these deficits have on daily life. Each condition is implemented with a texture, shader, or post-processing effect, and the conditions that are gaze-dependent track the user's eye movements in both AR and VR. This simulation suite had now been integrated as part of a physical diagnosis course for several semesters, and feedback from medical students has been generally positive, especially for understanding the effect of impairments on their future patients.

We hope that these simulations will give users a better understanding of the diseases themselves, improve empathy towards patients and loved ones, and assist with early self-diagnosis.

REFERENCES

- [1] D. Alexander, T. Nguyen, P. Keller, J. Orlosky, S. Brown, E. Wood, O. Ezenwoye, and W. Jirau-Rosalay. Design of visual deficit simulation for integration into a geriatric physical diagnosis course. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*, pp. 838–839. IEEE, 2020.
- [2] H. C. Ates, A. Fiannaca, and E. Folmer. Immersive simulation of visual impairments using a wearable see-through display. In *Proceedings of the ninth international conference on tangible, embedded, and embodied interaction*, pp. 225–228, 2015.
- [3] J. Lewis, D. Brown, W. Cranton, and R. Mason. Simulating visual impairments using the unreal engine 3 game engine. In *2011 IEEE 1st International Conference on Serious Games and Applications for Health (SeGAH)*, pp. 1–8. IEEE, 2011.
- [4] S. Patel, A. Pelletier-Bui, S. Smith, M. B. Roberts, H. Kilgannon, S. Trzeciak, and B. W. Roberts. Curricula for empathy and compassion training in medical education: a systematic review. *PloS one*, 14(8):e0221412, 2019.
- [5] G. Stewart and R. McCrindle. Visual impairment simulator for auditing and design. *Journal of Alternative Medical Research*, 9(4), 2017.